Goodness of Estimators

Learning Outcomes

- Rao-Blackwell Theorem
- MVUE
- MLE Properties

Rao-Blackwell Theorem

Let the joint distribution of $X_1, \dots, X_n$ depend on parameter θ and let T be a sufficient statistic for θ. If we consider estimating $h(\theta)$, a function of θ. If U is an unbiased estimator of $h(\theta)$, then the estimator $U^* = E(U|T)$ is also an unbiased estimator of $h(\theta)$ and has a variance no larger than U.

Minimum Variance Unbiased Estimator

Let $\mathcal{U}$ be a set of all unbiased estimators T of $\theta \in \Theta$ and $E(T^2) < \infty$. An estimator T_0 is said to be the minimum variance unbiased estimator for θ if

$$E\{(T_0 - \theta)^2\} \leq E\{(T - \theta)^2\}$$

for all $\theta \in \Theta$ and every $T \in \mathcal{U}$.

Invariance Property

If $\widehat{\theta}$ is an ML estimator of θ, then for any one-to-one function g, the ML estimator for $g(\theta)$ is $g(\widehat{\theta})$.

Large Sample Theory

Let $X_1, \ldots, X_n$ be a random sample from a distribution with parameter θ. Let $\widehat{\theta}$ be the MLE estimator for a parameter θ. As $n \to \infty$, then $\widehat{\theta}$ has a normal distribution with mean θ and variance $1/nI(\theta)$, where

$$I(\theta) = E\left[-\frac{\partial^2}{\partial\theta^2}\log\{f(X;\theta)\}\right]$$

Normal Distribution

Let $X_1, \ldots, X_n$ be a random sample with the following distribution:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$

Find the Information of μ.

Bernoulli Distribution

Let $X_1, \ldots, X_n$ be a random sample with the following distribution:

$$f(x) = p^x(1-p)^{1-x}$$

Find the Information of p.

Poisson Distribution

Let $X_1, \dots, X_n$ be a random sample with the following distribution:

$$f(x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

Find the Information of p.

Exponential Distribution

Let $X_1, \dots, X_n$ be a random sample with the following distribution:

$$f(x) = \frac{1}{\theta} e^{-x/\theta} \text{ for } x > 0$$

Find the Information of θ.